

Complex Analysis Exam Problems

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Fall 2025

Part I

1. Define $\gamma: [0, 2\pi] \rightarrow \mathbb{C}$ by

$$\gamma(t) = e^{it}.$$

Compute

$$\int_{\gamma} z^n dz$$

for every integer n .

Solution. Since

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi,$$

we have

$$\gamma'(t) = ie^{it}.$$

Therefore,

$$\int_{\gamma} z^n dz = \int_0^{2\pi} \gamma(t)^n \gamma'(t) dt = i \int_0^{2\pi} e^{i(n+1)t} dt.$$

If $n \neq -1$, then

$$\begin{aligned} \int_{\gamma} z^n dz &= i \left[\frac{e^{i(n+1)t}}{i(n+1)} \right]_0^{2\pi} \\ &= \frac{e^{2\pi i(n+1)} - 1}{n+1}. \end{aligned}$$

Because $n+1$ is an integer,

$$e^{2\pi i(n+1)} = 1,$$

and hence

$$\int_{\gamma} z^n dz = 0.$$

If $n = -1$, then

$$\int_{\gamma} z^{-1} dz = i \int_0^{2\pi} 1 dt = 2\pi i.$$

Thus

$$\int_{\gamma} z^n dz = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

■

2. Let γ be the closed polygon

$$[1 - i, 1 + i, -1 + i, -1 - i, 1 - i].$$

Find

$$\int_{\gamma} \frac{1}{z} dz.$$

3. 1. Show that a Möbius transformation has 0 and ∞ as its only fixed points if and only if it is a dilation, but not the identity.
2. Show that a Möbius transformation has ∞ as its only fixed point if and only if it is a translation, but not the identity.

Solution. 1. Let

$$T(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0.$$

Suppose that 0 and ∞ are the only fixed points of T . Since $T(0) = 0$, we must have $b = 0$. Since $T(\infty) = \infty$, we must have $c = 0$. Therefore,

$$T(z) = \frac{az}{d} = \lambda z,$$

where $\lambda = \frac{a}{d} \neq 0$. Thus T is a dilation. Moreover, $\lambda \neq 1$, since otherwise $T(z) = z$ would be the identity transformation, in which case every point would be fixed. Conversely, suppose that

$$T(z) = \lambda z, \quad \lambda \neq 0, 1.$$

Clearly, 0 and ∞ are fixed. If $z \in \mathbb{C}$ is any finite fixed point, then

$$\lambda z = z,$$

so

$$(\lambda - 1)z = 0.$$

Since $\lambda \neq 1$, it follows that $z = 0$. Hence the only fixed points of T are 0 and ∞ . Therefore, a Möbius transformation has 0 and ∞ as its only fixed points if and only if it is a dilation, but not the identity.

2. Suppose that ∞ is the only fixed point of T . Since $T(\infty) = \infty$, we must have $c = 0$. Thus

$$T(z) = \frac{az + b}{d} = \alpha z + \beta,$$

where $\alpha = \frac{a}{d} \neq 0$ and $\beta = \frac{b}{d}$.

If $\alpha \neq 1$, then a finite fixed point would satisfy

$$\alpha z + \beta = z,$$

and hence

$$z = -\frac{\beta}{\alpha - 1}.$$

This would give a finite fixed point, contradicting the assumption that ∞ is the only fixed point. Therefore, $\alpha = 1$, and so

$$T(z) = z + \beta.$$

Also, $\beta \neq 0$, since otherwise T would be the identity. Thus T is a translation, but not the identity.

Conversely, suppose that

$$T(z) = z + \beta, \quad \beta \neq 0.$$

Then ∞ is fixed. If $z \in \mathbb{C}$ were a finite fixed point, then

$$z + \beta = z,$$

which would imply $\beta = 0$, a contradiction. Hence ∞ is the only fixed point. Therefore, a Möbius transformation has ∞ as its only fixed point if and only if it is a translation, but not the identity. ■

4. Let G be a region, and suppose that $f: G \rightarrow \mathbb{C}$ is analytic such that $f(G)$ is a subset of a circle. Show that f is constant.

Solution. Suppose, for contradiction, that f is nonconstant. Since G is a region and $f: G \rightarrow \mathbb{C}$ is analytic and nonconstant, the Open Mapping Theorem implies that $f(G)$ is an open subset of $\mathbb{C}$. However, by hypothesis, $f(G)$ is contained in a circle. A circle contains no nonempty open subset of $\mathbb{C}$, since every open disk centered at a point on the circle contains points that are not on the circle. Thus $f(G)$ cannot be open. This is a contradiction. Therefore, f must be constant. ■

5. Let

$$f(z) = e^x \sin y + ie^x \cos y, \quad z = x + iy.$$

Determine whether f is analytic on the complex plane.

Solution. Write

$$f(z) = u(x, y) + iv(x, y),$$

where

$$u(x, y) = e^x \sin y \quad \text{and} \quad v(x, y) = e^x \cos y.$$

We compute the partial derivatives:

$$u_x = e^x \sin y, \quad u_y = e^x \cos y,$$

and

$$v_x = e^x \cos y, \quad v_y = -e^x \sin y.$$

For f to be analytic, the Cauchy–Riemann equations must hold:

$$u_x = v_y \quad \text{and} \quad u_y = -v_x.$$

The first equation gives

$$e^x \sin y = -e^x \sin y,$$

so

$$\sin y = 0.$$

The second equation gives

$$e^x \cos y = -e^x \cos y,$$

so

$$\cos y = 0.$$

However, there is no value of y for which both

$$\sin y = 0 \quad \text{and} \quad \cos y = 0.$$

Therefore, the Cauchy–Riemann equations do not hold at any point of $\mathbb{C}$. Hence f is nowhere analytic, and in particular it is not analytic on the complex plane. ■

6. Map

$$G = \mathbb{C} \setminus \{z : -9 \leq z \leq 9\}$$

onto the open unit disk by an analytic function f . Can f be one-to-one?

Solution. Let

$$G = \mathbb{C} \setminus [-9, 9].$$

We first construct an analytic map from G onto the open unit disk. Consider

$$h(z) = \frac{z - 9}{z + 9}.$$

For $z \in G$, the value $h(z)$ never lies on the nonpositive real axis. Hence we may take the principal square root and define

$$s(z) = \sqrt{\frac{z - 9}{z + 9}},$$

where the branch is chosen so that

$$\operatorname{Re} s(z) > 0.$$

Thus s maps G into the right half-plane. Now define

$$g(z) = \frac{s(z) - 1}{s(z) + 1}.$$

The Möbius transformation

$$w \mapsto \frac{w - 1}{w + 1}$$

maps the right half-plane onto the open unit disk. Also, $s(z) \neq 1$ for every finite z , since

$$\frac{z - 9}{z + 9} = 1$$

would imply $-9 = 9$. Therefore,

$$g(G) = \mathbb{D} \setminus \{0\},$$

where

$$\mathbb{D} = \{w : |w| < 1\}.$$

We now need to fill in the missing point 0. Let

$$\phi(w) = \frac{w - \frac{1}{2}}{1 - \frac{1}{2}w}.$$

This is an automorphism of $\mathbb{D}$, and since

$$\phi(0) = -\frac{1}{2},$$

we have

$$\phi(\mathbb{D} \setminus \{0\}) = \mathbb{D} \setminus \left\{-\frac{1}{2}\right\}.$$

Finally, define

$$f(z) = (\phi(g(z)))^2.$$

Explicitly,

$$f(z) = \left(\frac{g(z) - \frac{1}{2}}{1 - \frac{1}{2}g(z)} \right)^2, \quad g(z) = \frac{\sqrt{\frac{z-9}{z+9}} - 1}{\sqrt{\frac{z-9}{z+9}} + 1}.$$

This function is analytic on G . Moreover, the map $w \mapsto w^2$ maps

$$\mathbb{D} \setminus \left\{ -\frac{1}{2} \right\}$$

onto all of $\mathbb{D}$. Indeed, for any $\zeta \in \mathbb{D}$, choose a square root w of ζ . If $w = -\frac{1}{2}$, then the other square root $\frac{1}{2}$ is still in the domain. Hence

$$f(G) = \mathbb{D}.$$

We now show that no analytic map from G onto $\mathbb{D}$ can be one-to-one. The region G is not simply connected. For example, the curve

$$\gamma(t) = 10e^{it}, \quad 0 \leq t \leq 2\pi,$$

lies in G , and

$$\int_{\gamma} \frac{1}{z} dz = 2\pi i \neq 0.$$

Thus γ cannot be contracted to a point in G .

If an analytic map $f : G \rightarrow \mathbb{D}$ were both one-to-one and onto, then f would be a conformal equivalence between G and $\mathbb{D}$. In particular, G would be simply connected because $\mathbb{D}$ is simply connected. This is a contradiction. Therefore, there exists an analytic map from G onto the open unit disk, but such a map cannot be one-to-one. ■

7. Give the principal branch of

$$\sqrt{1 - z^3}.$$

Solution. The principal branch of the square root is defined by

$$\sqrt{w} = \exp\left(\frac{1}{2} \log w\right), \quad w \in \mathbb{C} \setminus (-\infty, 0],$$

where $\log$ denotes the principal logarithm.

Therefore, the principal branch of $\sqrt{1 - z^3}$ is

$$\sqrt{1 - z^3} = \exp\left(\frac{1}{2} \log(1 - z^3)\right).$$

Its domain consists of those z for which

$$1 - z^3 \notin (-\infty, 0].$$

Equivalently,

$$z^3 \notin [1, \infty).$$

Writing $z = re^{i\theta}$, the condition $z^3 \in [1, \infty)$ holds precisely when

$$r \geq 1 \quad \text{and} \quad \theta = \frac{2\pi k}{3}, \quad k = 0, 1, 2.$$

Thus the domain of the principal branch is

$$\mathbb{C} \setminus \bigcup_{k=0}^2 \{re^{2\pi i k/3} : r \geq 1\}.$$

Hence the three branch cuts are the rays beginning at the cube roots of unity $1, e^{2\pi i/3}, e^{4\pi i/3}$ and extending outward. This branch is characterized by

$$\sqrt[3]{1 - 0^3} = 1.$$

■

Part II

8. Compute the integral

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz,$$

where γ is the closed path

$$\gamma(t) = 3e^{it}, \quad 0 \leq t \leq 2\pi.$$

Solution. The path

$$\gamma(t) = 3e^{it}, \quad 0 \leq t \leq 2\pi,$$

is the positively oriented circle $|z| = 3$. Of the two singularities $z = 2$ and $z = 5$, only $z = 2$ lies inside γ . Set

$$h(z) = \frac{e^z}{(z-5)^5}.$$

Since $z = 5$ lies outside $|z| = 3$, the function h is analytic on and inside γ . Therefore,

by Cauchy's integral formula for derivatives,

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz = \int_{\gamma} \frac{h(z)}{(z-2)^2} dz = 2\pi i h'(2).$$

Now

$$\begin{aligned} h'(z) &= \frac{e^z}{(z-5)^5} - \frac{5e^z}{(z-5)^6} \\ &= \frac{e^z(z-10)}{(z-5)^6}. \end{aligned}$$

Thus

$$h'(2) = \frac{e^2(2-10)}{(2-5)^6} = -\frac{8e^2}{729}.$$

Consequently,

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz = -\frac{16\pi i e^2}{729}$$

■

9. Compute the integral in the previous exercise with γ the closed path

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

Solution. The path

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi,$$

is the positively oriented unit circle $|z| = 1$. The integrand

$$\frac{e^z}{(z-2)^2(z-5)^5}$$

has singularities only at $z = 2$ and $z = 5$. Since

$$|2| > 1 \quad \text{and} \quad |5| > 1,$$

both singularities lie outside γ . Therefore the integrand is analytic on and inside the unit circle. By Cauchy's theorem,

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz = 0$$

■